Summary

The Customer Churn Analysis project was conducted to identify the major factors influencing customer attrition and to derive actionable business insights using Exploratory Data Analysis (EDA) techniques in Python. The analysis was performed on a telecom customer dataset containing demographic details, subscription information, service usage patterns, and customer churn status.

Initially, the dataset was cleaned and preprocessed by handling missing values, verifying data types, removing inconsistencies, and checking for duplicate records to ensure data quality and accuracy before analysis. Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn were extensively used throughout the project for data manipulation and visualization.

The analysis revealed that approximately **26–27% of customers had churned**, while nearly **73–74% customers remained retained**, indicating that although the majority of customers stayed with the company, a significant proportion was still leaving the service. Pie charts and churn distribution plots clearly highlighted this imbalance.

Several comparative count plots, stacked bar charts, and categorical analysis visualizations were created to identify behavioral trends among churned customers. The analysis showed that customers using **Fiber Optic internet services experienced noticeably higher churn rates** compared to DSL users and customers without internet services. This indicates that service quality, pricing, or customer satisfaction issues may be influencing churn within this segment.

The study also identified that customers who lacked additional support services such as **OnlineSecurity, TechSupport, DeviceProtection, and OnlineBackup** were more likely to churn. In multiple charts, customers who subscribed to these value-added services demonstrated stronger retention rates, suggesting that customer support and security-related services play a crucial role in improving customer loyalty.

Contract type analysis further revealed that customers with **Month-to-Month contracts showed the highest churn percentage**, whereas customers with One-Year and Two-Year contracts had significantly lower churn rates. This suggests that long-term contractual commitments improve retention and reduce customer turnover.

Tenure-based analysis indicated that newer customers were more likely to churn compared to long-term customers. Churn percentages gradually decreased as customer tenure increased, showing that customers who remain with the company for longer durations tend to become more stable and loyal.

Additional visual analysis on categories such as **PhoneService, MultipleLines, StreamingTV, StreamingMovies, and SeniorCitizen status** helped identify customer groups with relatively higher churn tendencies. Senior citizens and customers without bundled support services displayed comparatively higher churn behavior.